Agriculture: Apple Pest Management Guidelines

Bactericide Efficacy—Conventional Products

Bactericide	Resistance risk (FRAC group number)[1]	Fire blight[4]		Phytotoxicity[6]	Plant growth regulator/SAR
		contact	systemic		
Actigard[5]	low (P 01)	0	2	0	2
Ag Streptomycin**, Agri-Mycin, Firewall	very high (25)	5	4	1	0
Apogee[4]	low (PGR)	0	2/3	0	3
Captan[2]	low (M 04)	3	0	0	0
Copper[3]	low (M 01)	3	0	2	0
Dithane, Manzate, Penncozeb[2]	low (M 03)	3	0	0	0
Kasumin	high (24)	5	5	1	0
Mycoshield, FireLine	high (41)	4	4	1	0

Rating: 5 = excellent and consistent, 4 = good and reliable, 3 = moderate and variable, 2 = limited and/or erratic, 1 = minimal and often ineffective, 0 = ineffective.

**	Not registered, label withdrawn or inactive in California.
1	Group numbers are assigned by the Fungicide Resistance Action Committee(http://www.frac.info/) (FRAC) according to different modes of action (for more information, see www.frac.info). Fungicides with different group numbers are suitable to alternate in a resistance management program. In California, make no more than one application of a fungicide with a mode-of-action group number associated with high resistance risk before rotating to a fungicide with a different mode-of-action group number; for other fungicides, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action group number.
2	These fungicides show some efficacy and should be used in mixtures with antibiotics as a component of resistance management programs. Captan is registered on apples, whereas Dithane is registered on apples and pears.
3	Though copper may be effective for scab and blight control under low disease pressure, copper products may cause fruit scarring or russeting.
4	Plant growth regulators (PGR) such as prohexadione calcium (Apogee) can be used in an integrated approach to reduce host susceptibility but do not have direct activity against fire blight.
5	Acibenzolar-S-methyl (FRAC P 01) is a host plant defense or systemic acquired resistance (SAR) inducer known to stimulate the salicylic acid pathway.
6	Higher numbers indicate higher phytotoxicity.

Acknowledgment: Adaskaveg et al., 2025. *Fungicides, Bactericides, Biocontrols, and Natural Products for Deciduous Tree Fruit and Nut, Citrus, Strawberry, and Vine Crops in California(/PDF/PMG/fungicideefficacytiming.pdf)*.

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